

0.1 重积分不等式证明

例题 0.1

(1) 设 $u \in C^1[0, 1]$ 满足 $u(0) = 0$, 证明:

$$|u(1)|^2 \leq \tanh 1 \cdot \left(\int_0^1 [|u(r)|^2 + |u'(r)|^2] dr \right).$$

(2) 设 $B \subset \mathbb{R}^3$ 是单位球, 证明:

$$\int_{\partial B} |f|^2 dS \leq \frac{e^2 - 1}{2} \left(\int_B |f|^2 dV + \int_B |\nabla f|^2 dV \right), \quad \forall f \in C^\infty(\bar{B}),$$

并证明系数不可改进.

注

(1) 第 (1) 问中, 为了分部积分能表示出 $u(1)$, 我们只需令 η 满足 $\eta'' = \eta, \eta'(1) = 1$.

(2) 第 (2) 问的思路是: 用极坐标将对球体的积分转化为径向部分的函数的积分.

证明 [Show/Hide Proof](#)

(1) 令 $\eta(r) \triangleq \frac{\sinh r}{\cosh 1}$, 则

$$\eta'(r) = \frac{\cosh r}{\cosh 1}, \quad \eta''(r) = \frac{\sinh r}{\cosh 1} = \eta(r).$$

由分部积分知

$$\int_0^1 u'(r)\eta'(r)dr = u(1)\eta'(1) - \int_0^1 u(r)\eta''(r)dr = u(1) - \int_0^1 u(r)\eta(r)dr.$$

从而

$$u(1) = \int_0^1 (u'(r)\eta'(r) + u(r)\eta(r))dr.$$

于是由 Cauchy 不等式和命题??可得

$$\begin{aligned} |u(1)|^2 &\leq \left| \int_0^1 (u'(r)\eta'(r) + u(r)\eta(r))dr \right|^2 \\ &\stackrel{\text{Cauchy 不等式 (离散版)}}{\leq} \left| \int_0^1 \sqrt{|u'(r)|^2 + |u(r)|^2} \cdot \sqrt{|\eta'(r)|^2 + |\eta(r)|^2} dr \right|^2 \\ &\stackrel{\text{Cauchy 积分不等式}}{\leq} \left(\int_0^1 (|u'(r)|^2 + |u(r)|^2) dr \right) \left(\int_0^1 (|\eta'(r)|^2 + |\eta(r)|^2) dr \right) \\ &= \left(\int_0^1 (|u'(r)|^2 + |u(r)|^2) dr \right) \int_0^1 \frac{\cosh^2 r + \sinh^2 r}{\cosh^2 1} dr \\ &= \left(\int_0^1 (|u'(r)|^2 + |u(r)|^2) dr \right) \int_0^1 \frac{\cosh(2r)}{\cosh^2 1} dr \\ &= \left(\int_0^1 (|u'(r)|^2 + |u(r)|^2) dr \right) \cdot \frac{\sinh 2}{2 \cosh^2 1} \\ &= \tanh 1 \cdot \int_0^1 (|u'(r)|^2 + |u(r)|^2) dr. \end{aligned}$$

(2) 对 $\forall w > 0$, 令 $F_w(r) \triangleq rf(rw)$ ($\forall r \in [0, 1]$), 则利用 (1) 可得

$$\begin{aligned} |f(w)|^2 &= |F_w(1)|^2 \leq \tanh 1 \int_0^1 (|F_w(r)|^2 + |F'_w(r)|^2) dr \\ &= \tanh 1 \int_0^1 \left(r^2 |f(rw)|^2 + \left| f(rw) + r \frac{\partial f(rw)}{\partial r} \right|^2 \right) dr \\ &= \tanh 1 \int_0^1 \left[r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 + |f(rw)|^2 + 2r^2 f(rw) \frac{\partial f(rw)}{\partial r} \right] dr \end{aligned}$$

$$\begin{aligned}
&= \tanh 1 \int_0^1 \left[r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 + \frac{d(r|f(rw)|^2)}{dr} \right] dr \\
&= \tanh 1 \int_0^1 \left(r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 \right) dr + \tanh 1 |f(w)|^2.
\end{aligned}$$

从而

$$\begin{aligned}
|f(w)| &\leq \frac{\tanh 1}{1 - \tanh 1} \int_0^1 \left(r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 \right) dr \\
&= \frac{e^2 - 1}{2} \int_0^1 \left(r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 \right) dr.
\end{aligned}$$

于是

$$\begin{aligned}
\int_{\partial B} |f(w)|^2 dS(w) &\leq \frac{e^2 - 1}{2} \int_{\partial B} \int_0^1 \left(r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 \right) dr dS(w) \\
&= \frac{e^2 - 1}{2} \int_0^1 \int_{|w|=1} \left(r^2 |f(rw)|^2 + r^2 |w \cdot \nabla f(rw)|^2 \right) dS(w) dr \\
&\stackrel{\text{第一类曲面积分换元法}}{=} \frac{e^2 - 1}{2} \int_0^1 \int_{|v|=r} \left(|f(v)|^2 + \left| \frac{v}{r} \cdot \nabla f(v) \right|^2 \right) dS(v) dr \\
&= \frac{e^2 - 1}{2} \int_0^1 \int_{|v|=r} \left(|f(v)|^2 + |\nabla f(v)|^2 \right) dS(v) dr \\
&\stackrel{\text{余面积公式}}{=} \frac{e^2 - 1}{2} \iiint_{0 \leq |v| \leq 1} \left(|f(v)|^2 + |\nabla f(v)|^2 \right) dV \\
&= \frac{e^2 - 1}{2} \int_B \left(|f|^2 + |\nabla f|^2 \right) dV.
\end{aligned}$$

特别地, 当

$$f(x, y, z) = \frac{\sinh \sqrt{x^2 + y^2 + z^2}}{\sqrt{x^2 + y^2 + z^2}}$$

时, 不等式等号成立.

□